

Element E

We used several mathematical and scientific principles in our engineering process. The most essential to the project was gravity. Gravity is what was intended to pull the petals down, not the motors. The motors were actually only used to hold the petals in place at the top of the rose and to slow the fall of the petals. We needed the force of gravity on the petals to outweigh the frictional forces between the fishing wire and the stem when the motors release the wire so that the petals could accelerate downwards. However, we wanted the petals to be light enough to create the falling petal effect and to make the drop slower and more dramatic. Weight plays a dominant role in gravitational force according to the equation $F=ma$ where a is the gravitational acceleration. Because of this, we balanced the weight of the petal by including small masses in the fabric while still using a relatively lightweight material. These mechanical principles were reviewed by field experts from both Duke that had either recently graduated or were currently participating in their graduate program studies and from theater experts. They gave us professional feedback that we recorded and took into account while constructing our rose such as the following points:

- “make the flowers fall pretty. Don’t let them lose their look for function” -stage manager
- “Buttons are hard to push for the stage actors. Especially if it’s the Beast and he’s wearing gloves” -another stage manager
- “explore other options to not use motors” -a four year graduate named Piper
- a remote is very difficult and intricate so it would be better to find other solutions
- use three motors, one for each scene -Piper
- make carves into dowels to lengthen string -Piper

In order to make the petals fall eloquently we schemed to adjust the motor unspool rate and speculated the effects of adding fans to blow the falling petals. As for the buttons, we had decided to send an engineering student to operate the controls as opposed to the actors. We also considered other options besides motors, the idea of having a motor for each scene, and carving into the dowels to lengthen string, none of which we implemented in our final plan.